

Table 4. Finite-time baseline results

Panel A. Finite-time baseline results

Initial state definition	Initial doublet probability	Doublet reaction yield (per encounter)	Quartet reaction yield (per encounter)	Primary superoxide (radical eq. per encounter)	Escape yield (per encounter)	Unresolved survival probability	Reaction + escape + survival
unpolarized I6/6	0.33333	0.184829	0.057300	0.242129	0.757828	0.000043	1.000000
doublet-manifold mixture PD/2	1	0.433386	0.012110	0.445496	0.554487	0.000017	1.000000
quartet-manifold mixture PQ/4	0 (roundoff)	0.060551	0.079895	0.140445	0.859498	0.000057	1.000000
manifold mixture pD=0.25	0.25	0.153759	0.062949	0.216708	0.783245	0.000047	1.000000
manifold mixture pD=0.75	0.75	0.340177	0.029056	0.369233	0.630740	0.000027	1.000000

The five initial-state benchmarks use the declared baseline sensitivity scenario at $t = 8/k_{ref}$. Unresolved survival is reported explicitly; these finite-time values are not asymptotic yields. Presentation values are rounded. All 5 source rows and all 43 source columns remain in table04_baseline_results.csv; 35 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.